STATE OF AI RESEARCH

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ABSTRACT

This study analyzes over 4 million academic papers that discuss AI in their abstracts, published between 2013 and mid-2026. The analysis measures keyword frequency, temporal trajectories, growth rates, citation distributions, geographic concentration, institutional output, open access rates, and title-versus-abstract coverage gaps. Key findings include the continued lead of neural networks by abstract-search volume (about 28% of corpus size), the 33.0x growth of large language model abstract mentions from 2018 to 2025, China exceeding the United States in total AI research volume by paper count, and nearly half of all papers receiving zero citations.

1 INTRODUCTION

In 2025, over 788,000 academic papers in the OpenAlex database mentioned artificial intelligence methods in their abstracts. Through mid-2026, 597,899 such papers have been recorded, putting the full year on pace for approximately 1.2 million total papers.

Many bibliometric studies of AI research rely on subject classification tags or curated keyword lists that may not cover cross-disciplinary usage of AI methods. When keyword search is used, title-based approaches include only papers where the author chose to place the method name in the title. A paper titled "Predicting Protein Stability Under Thermal Stress" that uses a neural network throughout its methods section would be invisible to a title-only search for "neural network." Abstract-level analysis addresses this gap by searching the text where authors describe their methods, results, and contributions.

To analyze these trends, a bibliometric corpus of 4,082,450 publications was defined by querying the OpenAlex scholarly database for academic papers published between 2013 and mid-2026 that explicitly mention AI-related terms in their abstracts. No papers were downloaded; all analyses were performed through API count queries. Data was retrieved on 8 June 2026.

The analysis covers 14 annual cohorts (2013-2026), measuring publication volume, n-gram frequency, growth rates, citation distributions, geographic output, and open access rates, and compares abstract-level search against title-only search for selected keywords.

Section 2 describes the corpus definition and analysis methods. Section 3 presents results across nine dimensions. Section 4 discusses the findings. Section 5 reviews related work. Section 6 concludes.

2 METHODOLOGY

2.1 DATA SOURCE

OpenAlex is an open scholarly database indexing over 250 million academic works [1; 19]. It provides free API access with structured metadata including titles, abstracts (stored as inverted index), author affiliations, citation counts, open access status, and machine-learned concept tags. OpenAlex was chosen over Web of Science or Scopus because it is freely accessible, covers preprints (including arXiv), and exposes structured API filters that allow exact-count queries without downloading raw data.

2.2 Corpus Definition

The corpus was defined using OpenAlex's `abstract.search` filter. The search query combined 10 AI-related terms using boolean OR logic.

Search terms. *artificial intelligence, machine learning, deep learning, neural network, language model, reinforcement learning, computer vision, natural language, generative, autonomous.*

Date range. 2013-01-01 through 2026-06-08.

Resulting corpus. 4,082,450 papers whose OpenAlex document type is journal article or preprint and that match the search filter (Table 1). Other document types were excluded from this analytical corpus.

Eight of the ten search terms are specific to AI. Two terms ("generative" and "autonomous") are broader and match non-AI papers. "Generative" can match generative grammar in linguistics. "Autonomous" can match autonomous systems in biology. Some fraction of the corpus consists of papers where the matched term refers to a non-AI context. This tradeoff was accepted because excluding these terms would miss entire AI research areas such as generative adversarial networks [10] (67,638 papers) and autonomous driving (71,930 papers).

Document Type	Count	Share	Source Type	Count	Share
Journal article	3,270,717	65.4%	Journal	2,232,178	44.6%
Preprint	668,948	13.4%	Repository (arXiv, SSRN, etc.)	1,370,590	27.4%
Book chapter	386,449	7.7%	Book series	246,842	4.9%
Dataset	298,570	6.0%	Conference proceedings	111,398	2.2%
Review	94,567	1.9%	eBook platform	69,676	1.4%
Dissertation	92,581	1.9%	Other / unclassified	973,099	19.5%
Other	191,951	3.8%			
Total	**4,082,450**				

Table 1: Corpus composition by document type and source type.

Repositories (primarily arXiv) account for 27.4% of the corpus, which is expected given how heavily AI researchers rely on preprints. Journals remain the largest single source at 44.6%.

2.3 Analysis Methods

All analyses were performed through direct OpenAlex API calls. No local text processing was applied.

1. **Keyword frequency.** For each keyword, bigram, or trigram of interest, a single API call was issued using `abstract.search:<term>,publication_year:2013-2026` and the `meta.count` field from the response was recorded.
2. **Growth detection.** For each keyword, the total publication count from 2025-2026 was divided by the total from 2022-2023 to produce a growth ratio.
3. **Time-series trajectories.** For 10 selected terms, year-by-year API calls were issued to obtain annual publication counts from 2013 (or the method's introduction year) through 2026.
4. **Citation distribution.** The `cited_by_count` filter was used to count papers in seven citation ranges (0, 1-10, 11-50, 51-100, 101-500, 501-1000, 1000+).
5. **Geographic and institutional analysis.** OpenAlex's `group_by` aggregation on `authorships.countries` and `authorships.institutions.lineage` was used to rank countries and institutions. A single paper with co-authors from multiple countries is counted once per country.
6. **Title vs. abstract comparison.** For 12 keywords, parallel queries were run using `title.search` and `abstract.search` and the ratio of abstract to title counts was computed.

2.4 Stemming and Precision

OpenAlex's search filters apply stemming, meaning a search for "agentic" also matches "agent" and "agents." For multi-word phrases ("retrieval augmented generation," "graph neural network") and proper nouns ("DeepSeek," "Claude," "Mistral"), stemming has minimal effect. For single common words ("diffusion," "safety," "clinical"), stemming can inflate counts by matching non-AI uses of the word. For example, searching "clinical" across all OpenAlex papers (2013-2026) returns 6,704,069 abstract matches, far exceeding the 4,082,450 papers in the AI-filtered corpus. Similarly, "agent" returns 2,179,789 matches, inflated by non-AI uses of the word in chemistry, pharmacology, and other fields. This limitation is discussed further in Section 4.5.

3 Results

3.1 Publication Volume

Year	Papers (Count)	YoY Growth	Cumulative Count	Share of Corpus
2013	93,226	-	93,226	1.9%
2014	97,510	+4.6%	190,736	1.9%
2015	105,609	+8.3%	296,345	2.1%
2016	115,423	+9.3%	411,768	2.3%
2017	137,237	+18.9%	549,005	2.7%
2018	185,192	+34.9%	734,197	3.7%
2019	242,286	+30.8%	976,483	4.8%
2020	305,903	+26.3%	1,282,386	6.1%
2021	369,519	+20.8%	1,651,905	7.4%
2022	411,098	+11.2%	2,063,003	8.2%
2023	520,861	+26.7%	2,583,864	10.4%
2024	662,417	+27.2%	3,246,281	13.2%
2025	788,012	+42.6%	4,190,811	18.9%
2026 (up to June)	597,899	-	4,082,450	16.2%

Table 2: Annual publication volume (abstract-level corpus).

The corpus grew from 93,226 papers in 2013 to 788,012 in 2025, a 10.1x increase over 12 years. The growth curve shows three distinct phases.

Phase 1 (2013-2016), slow growth, 4.6-9.3% per year. Annual output grew from 93,226 to 115,423 papers. Deep learning was still an active research area rather than a standard tool.

Phase 2 (2017-2022), deep learning adoption, 11.2-34.9% per year. Growth was fastest in 2017-2018 (+18.9% and +34.9%). The transformer architecture was introduced in 2017 [6], and deep learning frameworks (TensorFlow, PyTorch [20]) reached stable releases during this period. Growth decelerated to 11.2% in 2022.

Phase 3 (2023-2026), LLM-driven re-acceleration, 26.7-42.6% per year. Starting in 2023, growth re-accelerated sharply. The 2025 output (788,012) represents a 42.6% increase over 2024, the highest annual growth rate since 2018. ChatGPT was released in November 2022, and the subsequent expansion of LLM-related research (Section 3.3) likely contributed to this re-acceleration.

The 2026 cohort (597,899 papers recorded through early June) is on pace to reach approximately 1.2 million publications for the full year based on linear extrapolation. If realized, this would be the first year in which annual output in this corpus exceeds 1 million papers.

The corpus is heavily weighted toward recent years. Papers published from 2023 onward account for 58.8% of the total, meaning that the majority of AI-related academic work indexed by OpenAlex is less than four years old.

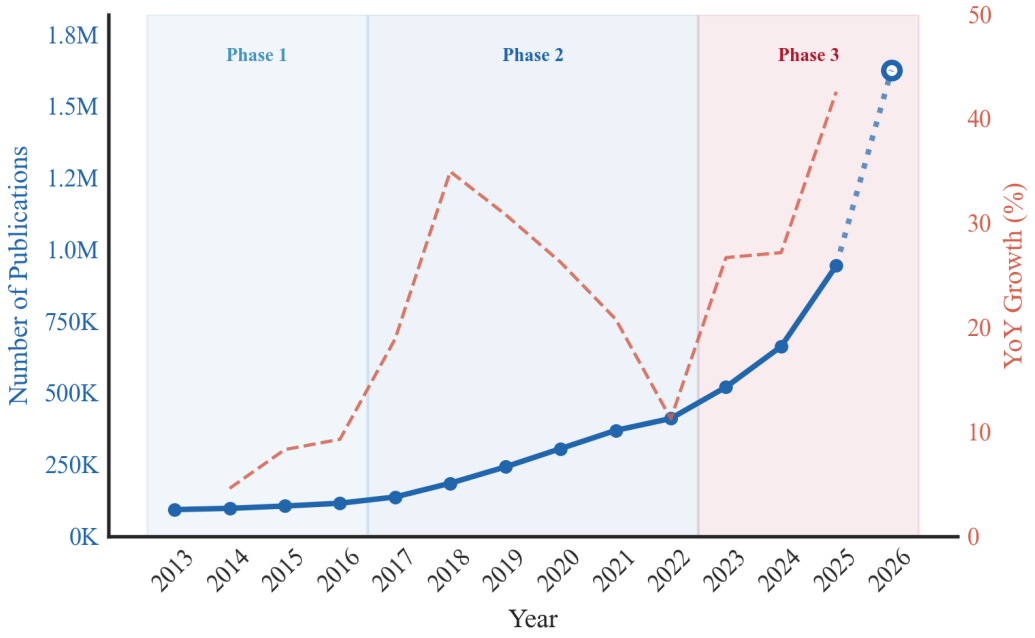


Figure 1: AI research publication volume, 2013-2026. The 2026 value is an annualized estimate based on January-June data (open marker).

3.2 N-GRAM FREQUENCY

Rank	Bigram	Mentions	Trigram	Mentions
1	neural network	1,147,683	deep neural network	518,431
2	machine learning	1,287,123	convolutional neural network	394,934
3	deep learning	980,070	large language model	292,873
4	artificial intelligence	745,358	artificial neural network	261,355
5	attention mechanism	432,079	support vector machine	239,347
6	large language	405,166	natural language processing	172,355
7	image classification	390,138	long short-term memory	137,359
8	recommendation system	387,638	recurrent neural network	88,266
9	medical imaging	359,104	graph neural network	86,453
10	feature extraction	256,159	random forest classifier	73,385

Table 3: Top 10 bigrams and trigrams in abstracts.

"Neural network" (1,147,683) dominates the list, appearing in 28.1% of all paper abstracts. "Machine learning" (1,287,123) and "deep learning" (980,070) round out the top three. Together, these three terms account for over 3.8 million abstract mentions in total (including duplicate counts where multiple terms appear in the same abstract).

"Attention mechanism" (432,079) ranks 5th. Attention-based architectures appear across NLP, computer vision, and multimodal tasks in this corpus, consistent with the transformer's influence since 2017 [6].

"Large language" (405,166) at rank 6 corresponds to the LLM growth in this corpus. It already surpasses older application-oriented terms such as "image classification" (390,138), "recommendation system" (387,638), and "medical imaging" (359,104).

"Feature extraction" (256,159) at rank 10 is a general-purpose term used across many machine learning pipelines.

"Deep neural network" (518,431) leads the trigrams, followed by "convolutional neural network" (394,934) [3]. Together these account for over 913,000 abstract mentions.

"Large language model" (292,873) at rank 3 has overtaken "artificial neural network" (261,355) and "support vector machine" (239,347). LLMs are discussed more broadly than title-only searches suggest (292,873 abstract mentions vs. 71,469 title mentions, a 4.1x ratio per Table 6).

"Support vector machine" (239,347) and "random forest classifier" (73,385) persist in the top 10. These classical methods continue to be widely referenced in abstracts, often as baselines for comparison or in application domains where simpler models remain competitive.

Both the bigram and trigram lists are top-heavy. The highest-ranked bigram (1,147,683) has 5.9x the count of the tenth (256,159), and the highest trigram (518,431) has 7.1x the count of the tenth (73,385). Five of the ten trigrams end in "network" (deep neural, convolutional neural, artificial neural, recurrent neural, graph neural), reflecting how many research areas have developed their own neural network variant. The bigram list splits between method-level terms in the top six (neural network through large language, totaling 5.37 million mentions) and application-level terms in the bottom four (image classification through feature extraction, totaling 1.39 million).

3.3 Time-Series Trajectories

Foundational terms. "Neural network" grew steadily from 23,395 abstract mentions in 2013 to 207,140 in 2025 (representing an 8.9x increase). "Deep learning" grew from 4,120 in 2013 to 216,713 in 2025 (a 52.6x increase). In 2025, "deep learning" surpassed "neural network" in annual abstract mentions for the first time (216,713 vs. 207,140), after closing the gap steadily from a 0.18 ratio in 2013 to 0.90 in 2024. "Reinforcement learning" grew from 1,784 in 2013 to 47,498 in 2025 (a 26.6x increase). "Transformer" grew from 7,201 abstract mentions in 2017 to 78,135 in 2025 (a 10.9x increase). The early counts (2017-2018) likely include non-AI uses of "transformer" (e.g., electrical engineering).

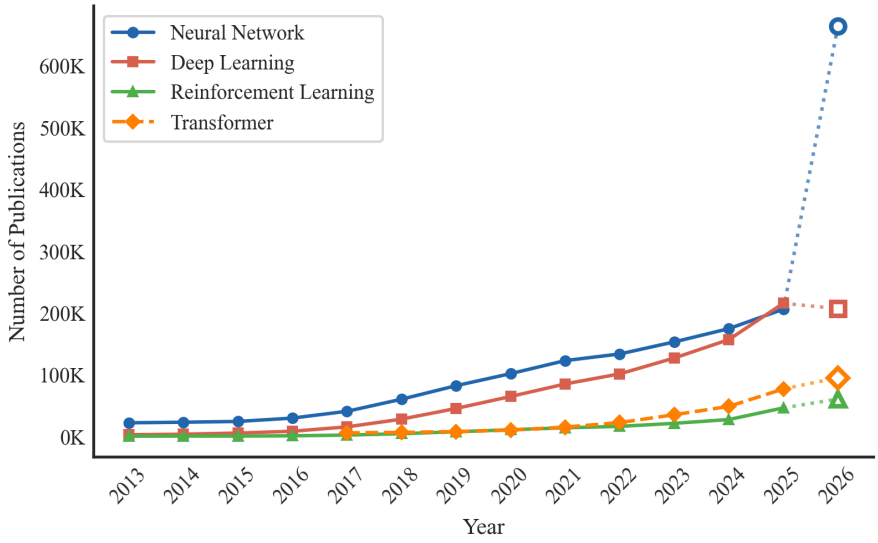


Figure 2: Trajectories of foundational AI terms by annual abstract mentions, 2013-2026. The 2026 values are annualized estimates based on January-June data (open markers).

The LLM trajectory. "Large language model" abstract mentions grew from 2,478 in 2018 to 81,894 in 2025, representing a 33.0x increase. The growth curve has a clear inflection point. Between 2018 and 2022, mentions grew at a modest pace (from 2,478 to 7,931 mentions, or a 2.4x increase over four years). Between 2022 and 2025, mentions grew 12.2x in three years. The acceleration is visible year by year, jumping to 21,612 in 2023 (2.7x), 49,970 in 2024 (2.3x), and 81,894 in 2025 (1.9x). The annualized 2026 estimate (approximately 170,000 papers, based on 84,957 recorded through

June) suggests another 1.8x increase over the 2025 volume. By 2025, LLM papers accounted for 10.3% of all AI papers in the corpus, up from 4.1% in 2023. No other term grew at a comparable rate over the same period. From 2022 to 2025, "transformer" grew 3.3x, "reinforcement learning" 2.7x, "deep learning" 2.1x, and "neural network" 1.5x.

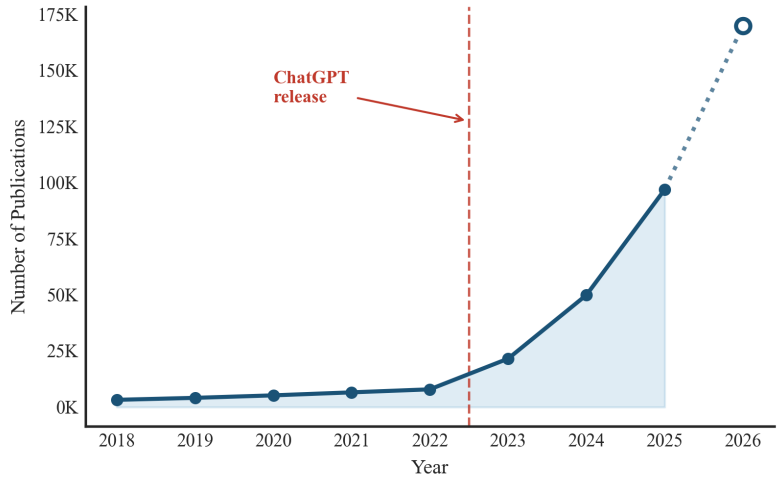


Figure 3: Growth of "large language model" in paper abstracts. The 2026 value is an annualized estimate based on January-June data (open marker).

In 2022, LLM papers were 1.9% of the corpus. By 2025, that share had grown to 10.3%. If the 2026 trajectory holds, roughly one in nine AI papers will mention large language models by year-end.

Rising terms. "Diffusion model" grew from 18,640 abstract mentions in 2019 to 49,862 in 2025 (a 2.7x increase). Early counts include non-AI uses of "diffusion model" (e.g., diffusion of innovations in social science). "Federated learning" [12] grew from 46 mentions in 2017 to 18,519 in 2025 (a 402.6x increase). "Graph neural" grew from 966 mentions in 2017 to 21,873 in 2025 (a 22.6x increase). "Knowledge graph" grew from 1,700 mentions in 2013 to 16,519 in 2025 (a 9.7x increase).

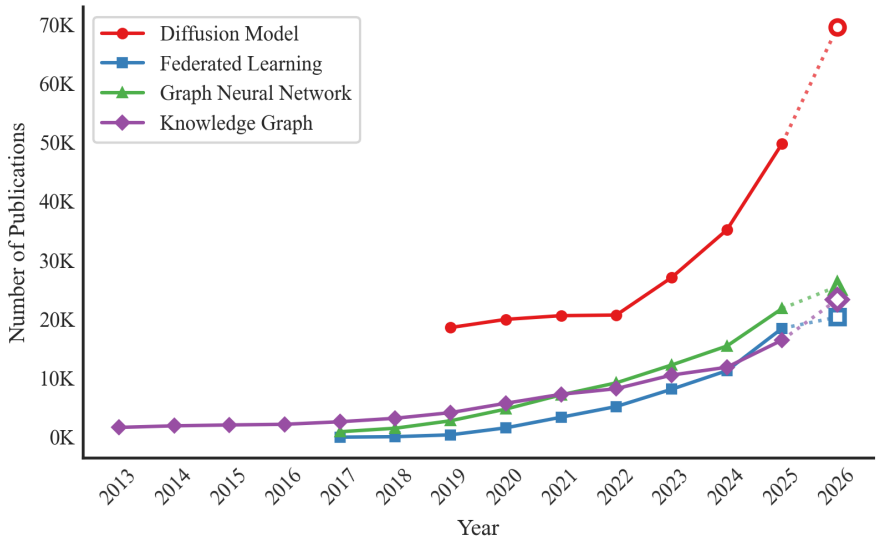


Figure 4: Rising AI terms by annual abstract mentions. The 2026 values are annualized estimates based on January-June data (open markers).

"Generative adversarial" grew from 6 papers in 2014 to 13,613 in 2025, totaling 67,638 papers across the corpus. Year-over-year growth peaked at 682% in 2017 and 189% in 2018, then fell to 4.5% in 2022 before settling at around 23% in 2024-2025. By comparison, "diffusion model" reached 49,862 annual mentions in 2025, 3.7x the GAN count that year. The lifecycle pattern is discussed in Section 4.3.

3.4 Fastest-Rising Keywords

Keyword	2025-2026 (Count)	2022-2023 (Count)	Growth (Ratio)
deepseek	11,033	13	785.6x
retrieval augmented generation	18,196	347	53.5x
jailbreak	2,803	110	27.6x
retrieval-augmented	21,105	1,101	19.2x
mistral	4,361	260	16.8x
llm	161,771	10,125	16.0x
copilot	5,699	356	16.0x
rag	19,193	1,250	15.4x
gemini	22,365	1,650	13.6x
guardrail	5,046	521	9.7x

Table 4: Top 10 fastest-rising keywords in paper abstracts (2025-2026 vs. 2022-2023).

The keyword growth data groups into three categories. Growth ratios were computed by dividing 2025-2026 abstract counts by 2022-2023 counts for each term (Section 2.3).

Category 1, named models as research subjects. "DeepSeek" (785.6x), "Mistral" (16.8x), and "Gemini" (13.6x) are all names of specific commercial models, not architectures or methods. Their presence in a fastest-rising keyword list suggests that researchers are spending more time evaluating and comparing named products.

Category 2, the RAG pipeline. "Retrieval augmented generation" (53.5x), "retrieval-augmented" (19.2x), and "RAG" (15.4x) all show rapid growth. Lewis et al. [8] introduced retrieval-augmented generation for knowledge-intensive NLP tasks, and the pattern has since been widely adopted. Combined, these three variants account for 58,494 papers in 2025-2026, making RAG the single largest area of growth in Table 4 by absolute volume after "LLM" itself.

Category 3, safety and reliability. "Jailbreak" (27.6x) and "guardrail" (9.7x) track the growing research effort to make language models reliable and safe. "Jailbreak" research (2,803 papers in 2025-2026, up from 110 in 2022-2023) investigates adversarial prompts that circumvent model safety filters. "Guardrail" (5,046, up from 521) covers techniques for constraining model outputs.

All ten of the fastest-rising keywords are related to large language models. None involve computer vision, reinforcement learning, or graph methods. "Copilot" (16.0x), the remaining term in the table, refers to LLM-based assistants for coding (GitHub Copilot), productivity (Microsoft 365 Copilot), and other domains.

Growth ratios from small baselines should be read with caution. "DeepSeek" (785.6x) had only 13 papers in the 2022-2023 baseline. By absolute count, "LLM" (161,771 papers in 2025-2026) is the largest term in the table, and the three RAG variants total 58,494. This table shows where research attention is shifting, not what contributes the most new papers overall. Established terms like "deep learning" added 58,766 papers from 2024 to 2025, far more than any single term in Table 4, but "deep learning" does not appear in the table because its growth ratio is low (approximately 2.1x over the same period) due to its large baseline.

3.5 Citation Distribution

The citation distribution is extremely right-skewed. Nearly half of all papers (48.9%) have received zero citations to date. Only 2,475 papers (0.05%) have accumulated more than 1,000 citations. Consequently, the median paper in this AI corpus has zero citations. This figure is partly inflated by recency, as papers published in 2024-2026 have had little time to accumulate citations.

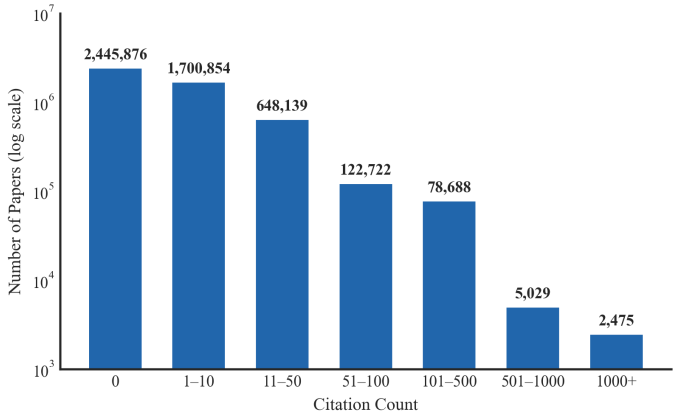


Figure 5: Citation distribution across the 4M+ AI papers in the corpus, shown on a log scale.

The most-cited paper is ResNet [2] with 221,202 citations, approximately 1.9x the next entry. Several of the most-cited papers in the corpus, including the DSM-5 [23] (113,579 citations) and the lme4 statistics package [24] (84,949), are not AI research contributions but mention related methods in their abstracts, illustrating the breadth of abstract-level search. Among AI-specific papers, the top entries (ResNet, Deep Learning [5], Adam [4], AlexNet, VGGNet, Faster R-CNN, XGBoost) are all widely reused building blocks. Papers that provide reusable infrastructure can accumulate orders of magnitude more citations than application-specific work, because they are cited by every downstream paper that builds on the tool or architecture, regardless of application domain.

3.6 Geographic Distribution

China leads with 826,032 publications, 29.8% higher than the United States (636,256). India ranks third with 369,931 publications, surpassing Japan (333,896) and the United Kingdom (216,177). The top two countries alone account for 42.5% of all country-paper assignments, and the top ten account for 82.5%. The gap between second and third place (1.94x, US to India) is larger than the gap between first and second (1.22x, China to US). The five largest European contributors (UK, Germany, Italy, France, Spain) combine to 664,851 country-paper assignments, comparable to the US alone (636,256) but well below China.

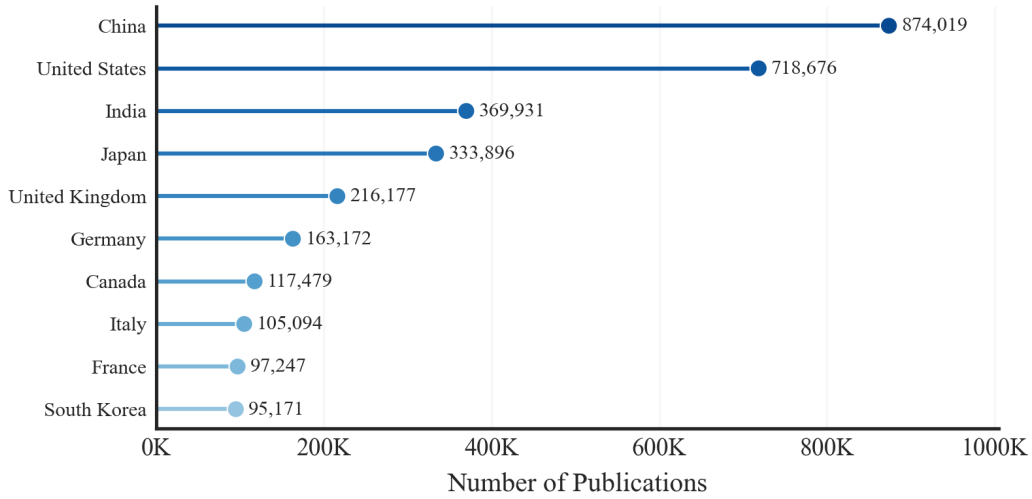


Figure 6: Top 10 countries by AI research output (4M+ paper abstract-level corpus, 2013-2026). A single paper with co-authors from multiple countries is counted once per country.

3.7 Institutional Output

Rank	Institution	Country	Papers (Count)	Share of Corpus
1	Chinese Academy of Sciences	China	74,921	1.50%
2	CNRS (French Natl. Research Centre)	France	50,145	1.00%
3	University of London	UK	34,887	0.70%
4	Tsinghua University	China	30,519	0.61%
5	Univ. of Chinese Academy of Sciences	China	23,911	0.48%
6	Shanghai Jiao Tong University	China	23,695	0.47%
7	Zhejiang University	China	23,246	0.46%
8	Harvard University	US	21,529	0.43%
9	US Department of Energy	US	20,244	0.40%
10	Peking University	China	20,127	0.40%

Table 5: Top 10 institutions by paper count.

The Chinese Academy of Sciences leads with 74,921 papers, 49.4% more than CNRS (50,145). Six of the top ten institutions are Chinese. Harvard (21,529) is the highest-ranked US institution at position 8.

OpenAlex's institution taxonomy includes umbrella organizations (CNRS, Helmholtz Association, US Department of Energy) alongside individual universities. These umbrella organizations aggregate papers from their constituent laboratories and institutes, which inflates their counts relative to standalone universities.

3.8 Open Access

Of the 4,082,450 papers in the corpus, 3,043,557 (60.8%) are published as open access (OA) literature, while 1,960,226 (39.2%) remain behind publisher paywalls. For context, Piwowar et al. [11] estimated the baseline open access rate across all academic fields at 28% in 2018. The higher rate in this corpus is likely due to the AI community's preprint culture, where arXiv is widely used to share work before formal publication.

3.9 Title vs. Abstract Comparison

Keyword	Title Search	Abstract Search	Abstract Only	Ratio
neural network	433,556	1,147,683	1,089,056	3.5x
machine learning	495,798	1,287,123	791,325	2.6x
deep learning	374,975	980,070	605,095	2.6x
large language model	71,469	292,873	221,404	4.1x
diffusion model	42,120	324,073	281,953	7.7x
transformer	129,524	316,216	186,692	2.4x
hallucination	10,711	48,759	38,048	4.6x
fairness	78,835	429,288	350,453	5.4x
retrieval augmented	8,137	27,394	19,257	3.4x
federated learning	39,481	59,298	19,817	1.5x
reinforcement learning	104,021	201,098	97,077	1.9x
knowledge graph	28,201	90,234	62,033	3.2x

Table 6: Title-only vs. abstract search for selected keywords.

The ratio of abstract-to-title matches varies from 1.5x ("federated learning") to 7.7x ("diffusion model"). "Diffusion model" (7.7x) is discussed in 324,073 abstracts but placed in the title of only 42,120 papers. However, as noted in Section 3.3, the abstract count for "diffusion model" includes non-AI uses (e.g., diffusion of innovations in social science), which inflates the ratio. The remaining high-ratio terms are more interpretable.

"Fairness" (5.4x) and "hallucination" (4.6x) also show high ratios. Both "fairness" and "hallucination" have significant non-AI uses. "Hallucination" appears in psychiatric and neuroscience

literature, and "fairness" in social science and economics. The abstract counts for these two terms are inflated by non-AI papers, similar to "diffusion model."

"Federated learning" (1.5x) has the lowest ratio, meaning papers that discuss federated learning almost always include it in their title. This suggests that federated learning is typically the main contribution of the paper, not a supporting technique.

For bibliometric work, these ratios suggest that title-only analysis may systematically undercount terms that are used as tools across disciplines, and abstract-level analysis returns more of that usage.

4 Discussion

4.1 The Persistence of Neural Networks

As shown in Section 3.2, "neural network" remains the most frequently referenced AI concept in the corpus, appearing in 28.1% of all abstracts. It holds this position despite public attention moving to large language models. The 3.5x ratio between abstract and title counts (Table 6) suggests that most papers using neural networks do not place the term in their titles. In this corpus, neural networks appear to function as a standard tool rather than a novel contribution. Growth has been steady across all 12 years in the dataset (Section 3.3).

4.2 The LLM Inflection Point

The LLM trajectory (Section 3.3) shows the sharpest inflection in the corpus. No other term matches this acceleration profile. The fastest-rising keywords (Table 4) suggest that research activity has moved from training LLMs to building systems around them. RAG (53.5x growth) is one example, as it focuses on integrating external knowledge into LLM pipelines. "Hallucination," "guardrail," and "jailbreak" indicate growing attention to reliability and safety.

The abbreviation "LLM" itself grew 16.0x (Table 4), which may indicate the abbreviation becoming as common as the full phrase in abstracts. As noted in Section 3.4, every one of the ten fastest-rising terms is related to large language models, and none involve computer vision, reinforcement learning, or graph methods.

4.3 Method Lifecycles

The time-series data (Section 3.3) places each value at a different lifecycle stage. Using the compound annual growth rate (CAGR) from 2022 to 2025 as the primary metric, with a corpus-wide baseline of 32.0%, Figure 7 maps 20 values onto an interpretive hype cycle.

Innovation trigger and ascending hype. Chain of thought prompting (64.6% CAGR), coined in 2022, is the newest term in the figure. Vision language models (83.9% CAGR, 2,582 to 16,067 papers) and zero-shot learning (68.1%) grew in parallel with LLM adoption. RAG grew from 85 papers in 2022 to 9,329 in 2025 (378.8% CAGR, 110x), and is placed near the peak accordingly.

Peak of inflated expectations. Large language models have the highest CAGR at 128.1% (7,931 to 81,894 papers). The annualized 2026 estimate (170,000 papers) suggests continued growth.

Trough of disillusionment. Vision transformers (60.9% CAGR) still show high growth but have moved from novelty to standard architecture since their 2021 peak. Diffusion models (33.9%) had their breakout in 2022-2023 and are now maturing. GANs (21.8%) and BERT (30.3%) have been largely superseded and sit deeper in the trough.

Slope of enlightenment. Six values fall in this zone, all above the corpus average: multimodal learning (52.9%), federated learning (52.3%), transformers (48.8%), reinforcement learning (39.2%), contrastive learning (35.8%), and graph neural networks (33.2%). All show steady adoption past their initial growth phase.

Plateau of productivity. Five values have below-average growth: deep learning (28.3% CAGR, 216,713 papers in 2025), knowledge graphs (25.9%), transfer learning (22.6%), CNNs (16.5%), and neural networks (15.4%). All continue to grow but at or below the corpus rate.

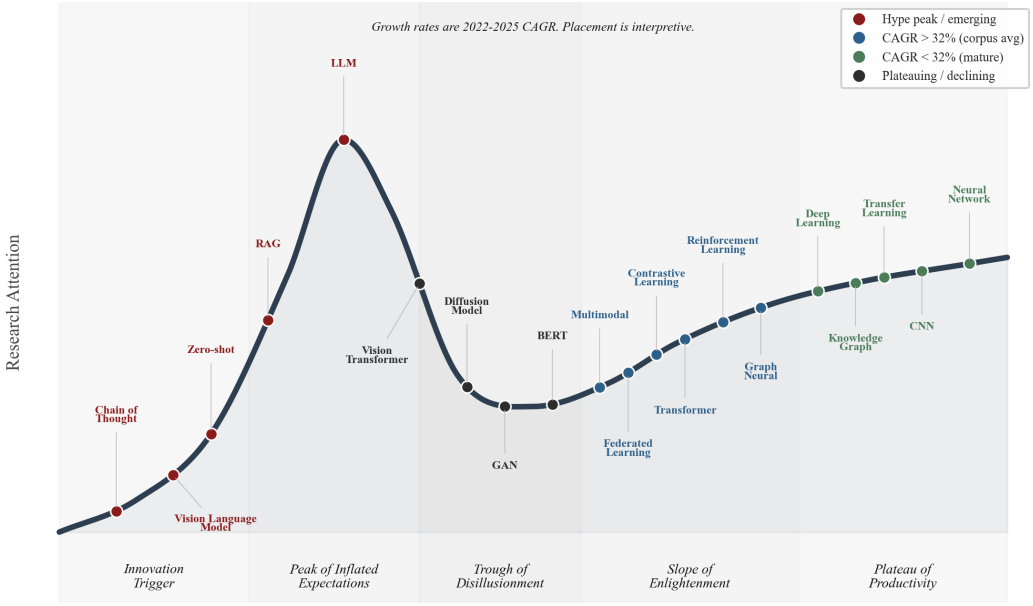


Figure 7: Interpretive hype cycle for AI research terms. Growth rates shown are 2022-2025 CAGR (corpus average 32.0%). See legend for color coding.

4.4 The China-US Research Balance

4.4.1 The Crossover

China and the United States started the decade at near-parity. In 2013, the US produced 13,829 AI-related papers (by abstract count) while China produced 12,074. Both countries grew steadily through 2018, but their trajectories diverged after that.

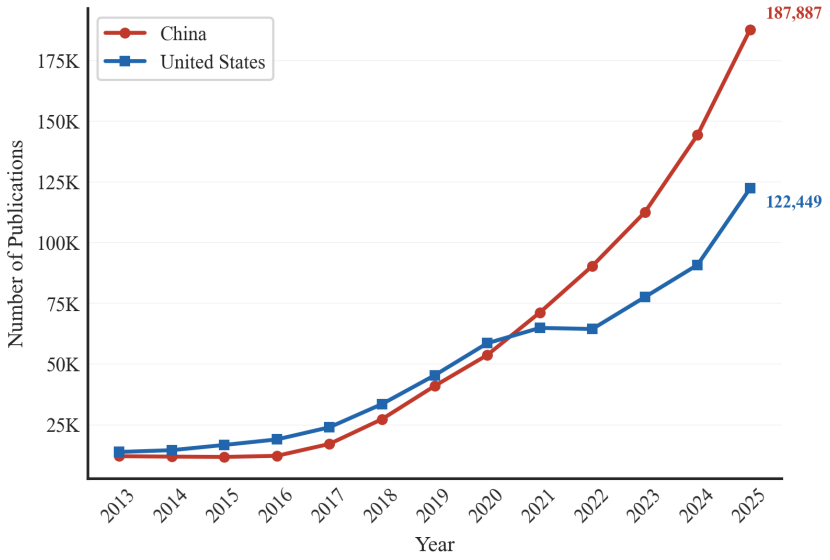


Figure 8: AI research output in this corpus for China and the United States, 2013-2025. China exceeded US output starting in 2021, with the difference increasing annually.

Year	China	United States	India	Japan	UK	China/US
2013	12,074	13,829	2,761	2,223	4,312	0.87
2014	11,897	14,551	3,445	2,277	4,557	0.82
2015	11,766	16,713	3,976	2,478	5,030	0.70
2016	12,229	18,988	4,996	2,913	5,762	0.64
2017	17,125	23,991	5,919	3,616	7,057	0.71
2018	27,353	33,567	8,592	5,075	9,567	0.81
2019	41,011	45,404	11,585	6,508	12,732	0.90
2020	53,743	58,622	17,612	7,712	16,574	0.92
2021	71,273	64,931	26,158	8,854	19,788	**1.10**
2022	90,485	64,486	35,760	9,420	19,798	**1.40**
2023	112,646	77,664	48,990	11,095	24,429	**1.45**
2024	144,452	90,915	65,398	12,657	28,174	**1.59**
2025	175,288	106,894	89,287	15,558	37,104	**1.53**

Table 7: Year-by-year AI research output for top 5 countries.

The crossover occurred in 2021. That year, China produced 71,273 papers while the US produced 64,931. US output fell slightly between 2021 and 2022 (64,931 to 64,486), while China continued to accelerate. By 2025, the gap had widened to 53.4% (175,288 vs. 106,894).

The US slowed from 2020 to 2022. In this corpus, US output grew 10.0% over two years (58,622 to 64,486), compared to 68.4% growth for China over the same period (53,743 to 90,485). Growth resumed in the US from 2023 onward (77,664 to 106,894 by 2025, a 57.7% increase over two years), but not fast enough to close the gap.

Paper counts measure research volume, not research impact. This study does not analyze citation-weighted metrics, shares of top-1% highly cited papers, or venue prestige, which may yield different rankings. Additionally, differing corpus definitions alter relative country ranks; for instance, while the Stanford AI Index 2026 report [9] restricts its analysis to computer science publications and lists India's share (7.55% in 2024) slightly ahead of the United States (7.29%), our broader cross-disciplinary corpus shows the United States retaining a clear lead over India (90,915 vs. 65,398 papers in 2024). Publication incentive structures also differ across countries. Institutional pressures (such as tenure requirements, h-index targets, and ranking criteria) have been noted in the literature for both China and the US, and may inflate output independently of research contribution.

4.4.2 Term-Specific Comparisons

The China-US gap is not uniform across terms.

Term	China	US	Combined	China/US
transformer	89,302	33,506	122,808	2.67x
federated learning	16,356	8,689	25,045	1.88x
neural network	332,858	177,752	510,610	1.87x
deep learning	241,216	140,965	382,181	1.71x
reinforcement learning	50,910	31,277	82,187	1.63x
diffusion model	66,832	56,315	123,147	1.19x
large language model	35,923	47,363	83,286	**0.76x**

Table 8: China vs. US paper counts by term (2013-2026, abstract search).

In the corpus, China leads in six of seven term categories. China's lead is strongest in "transformer" (2.67x), "federated learning" (1.88x), and "neural network" (1.87x). But the US leads in "large language model" (47,363 vs. 35,923, or 1.32x the Chinese count). While China produces more AI papers overall in the sample, the US produces more papers on the fastest-growing category since 2023.

The term-level shares within each country differ. "Neural network" appears in 38.1% of Chinese AI papers but 24.7% of US papers. "Transformer" appears in 10.2% of Chinese papers but only 4.7% of US papers. By contrast, "diffusion model" is nearly even as a share of each country's output (7.6%

for China, 7.8% for the US), and "large language model" tilts toward the US (6.6% vs. 4.1%). The overall pattern is that China's volume lead is concentrated in established neural network methods, while the US has a proportionally larger share in LLM and diffusion model research.

4.4.3 The LLM Convergence

The US led China in LLM research output throughout 2020-2024, with the gap narrowing overall, though it temporarily widened in 2023 before resuming its convergence. In 2020, the US produced 2.4x more LLM papers than China. By 2025, China reached parity (China: 15,008 vs. US: 14,735, a ratio of 1.02). Chinese LLMs such as DeepSeek-V3 [7], Qwen, and Yi were released during the same period, giving Chinese researchers domestic foundation models to study, benchmark, and extend.

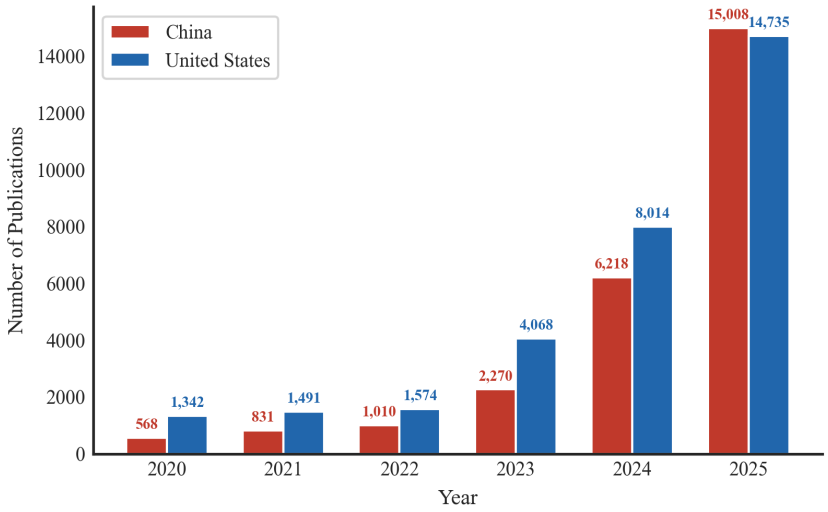


Figure 9: LLM research papers per year for China and the United States. The US led from 2020-2024, with China reaching parity in 2025.

From 2020 to 2022, the US produced more LLM papers than China, with the ratio narrowing from 2.36x to 1.56x. In 2023, the gap temporarily widened (to 1.79x) as US output surged following the release of ChatGPT. By 2024, the gap had narrowed again to 1.29x. In 2025, China slightly exceeded the US for the first time (15,008 vs. 14,735).

4.4.4 India's Acceleration

India shows a distinct growth pattern. In the corpus, Indian AI research output grew from 2,761 papers in 2013 to 89,287 in 2025, a 32.3x increase. This growth multiple is higher than both China (15.6x) and the US (8.9x) over the same period. India produced 89,287 AI papers in 2025, more than Japan (15,558) and the UK (37,104) combined.

India's growth rate has been accelerating. Between 2013 and 2018, Indian output grew 3.1x. Between 2018 and 2025, it grew 10.4x.

4.5 Limitations

Stemming and noise. OpenAlex applies stemming to search queries, which inflates counts for common words. "Explainable" returns 2,544,915 abstract matches, which is clearly inflated by stemming matching "explain" in non-AI contexts. Single-word keyword counts in Table 3 are best interpreted with this caveat in mind. Multi-word phrases and proper nouns are minimally affected.

Cross-disciplinary noise. As noted in Section 2.2, some fraction of the corpus consists of non-AI papers matching broad terms like "autonomous" or "generative." This is an inherent tradeoff of abstract-level analysis versus title-level analysis (which is more precise but less complete).

Temporal coverage. The 2026 cohort covers January through early June. Annualized projections assume even distribution throughout the year, which may not hold due to conference deadlines and journal publication cycles.

Multi-counting. A paper co-authored by researchers in China and the United States is counted once in each country's total. Country-level paper counts therefore sum to more than the corpus total. The same applies to institutions.

OpenAlex coverage. OpenAlex indexes over 250 million works but does not cover all academic literature. Non-English publications, conference proceedings from smaller venues, and technical reports may be underrepresented. OpenAlex has stronger coverage of English-language journals and repositories.

Causation vs. correlation. Growth in keyword frequency tracks research attention, not research quality or real-world deployment. A 16x increase in "LLM" papers does not mean LLMs are 16x more useful than they were in 2022.

Volume vs. impact. All country and institutional comparisons are based on paper counts, which measure volume, not impact. The caveats discussed in Section 4.4.1 (citation-weighted metrics, venue prestige, and differing publication incentive structures) apply to all geographic and institutional comparisons in this paper.

Cross-disciplinary inclusion. As noted in Section 3.5, the most-cited papers in the corpus include non-AI entries (e.g., DSM-5, lme4) whose abstracts happen to mention AI-related terms. This is an inherent feature of abstract-level search.

No normalization against baseline. The growth rates in this paper are raw, not normalized against overall academic publishing growth. Some of the observed AI growth is due to baseline expansion of scholarly output rather than AI-specific acceleration.

5 Related Work

5.1 AI Bibliometric Studies

The Stanford HAI AI Index Report [9] is a widely cited annual survey of AI research trends. The 2026 edition tracks publications, patents, investment, and policy across multiple data sources including Dimensions, Epoch, and LMSYS. This study differs in two ways. First, it uses abstract-level search rather than subject classification, capturing cross-disciplinary AI method usage. Second, every count in the analysis can be verified through the free OpenAlex API, rather than requiring access to proprietary databases.

The AI Index 2021 report [14] found that China and the US together dominated global AI publication output. In this corpus, China (826,032) and the US (636,256) together total 1,592,695 paper-country assignments (31.8% of the corpus total, though this figure is inflated by co-authored papers counted in both countries). The lower relative share compared to the AI Index is consistent with this corpus including a broader set of AI-adjacent documents beyond the core AI fields tracked by that report.

Jurowetzki et al. (2021) documented increasing overlap between academic and commercial AI development [15]. The growth of named models (DeepSeek, Gemini, Mistral) as the fastest-rising research terms in this corpus may be consistent with that trend. Ahmed and Wahed (2020) examined the compute divide between industry and academic labs [18]. The title vs. abstract comparison (Table 6) adds a methodological datapoint, showing that abstract search returns 1.5x to 7.7x more papers per keyword than title-only search, consistent with standard recommendations for thorough search strategies [22].

5.2 Compute and Scaling Studies

Sevilla et al. (2022) analyzed compute trends in machine learning, documenting distinct scaling eras with doubling times ranging from 5-6 months in the deep learning era to approximately 10 months in the large-scale era [13]. The periods of fastest publication growth in this corpus (2017-2018 and 2023-2025) align with the periods when compute scaling enabled new model capabilities.

Hoffmann et al. (2022) introduced compute-optimal scaling laws ("Chinchilla scaling") [16], and Kaplan et al. (2020) characterized neural scaling laws [17]. The growth of "large language model" (33.0x from 2018 to 2025) tracks the research activity that followed, accelerated by models like GPT-3 [21].

6 Conclusion

Three broad patterns emerge from this corpus of five million papers. Established terms (neural networks, deep learning, reinforcement learning) continue to dominate in this corpus by accumulated volume. The LLM category has grown faster than any other term (33.0x over seven years). And research on reliability and safety (hallucination, guardrail, jailbreak) has grown rapidly alongside increased deployment.

Eight findings from this analysis.

1. **Neural networks remain dominant.** "Neural network" appears in 1,147,683 paper abstracts (28.1% of the corpus). In 2025, "deep learning" surpassed "neural network" in annual abstract mentions for the first time, though "neural network" retains a larger cumulative total.
2. **LLMs are the fastest-growing category.** "Large language model" grew from 2,478 abstracts in 2018 to 81,894 in 2025 (33.0x), with an inflection point in 2022-2023 coinciding with the release of ChatGPT. By 2025, LLM papers accounted for 10.3% of the corpus, up from 1.9% in 2022.
3. **Research attention appears to be shifting from architecture toward application.** The fastest-rising terms are not architectures but patterns (RAG, 53.5x), safety concepts (jailbreak, 27.6x), and specific model names (DeepSeek, 785.6x). All ten of the fastest-rising keywords in Table 4 are related to large language models.
4. **China leads in volume.** In the corpus, China produces 21.6% more AI research papers than the United States by paper count. China leads in six of seven term categories, but the US leads in "large language model" (1.32x the Chinese count). China reached LLM parity with the US in 2025. This study does not measure citation impact or venue prestige, which may yield different rankings.
5. **Almost half of all papers go uncited.** 48.9% of papers have zero citations, while only 2,475 papers (0.05%) have exceeded 1,000 citations. This figure is inflated by recent publications that have not yet had time to accumulate citations.
6. **Title-only analysis misses most AI research.** Abstract search returns 1.5x to 7.7x more papers per keyword, depending on the term. The abstract-to-title ratio is smallest for methods that are typically the main contribution of a paper (federated learning, 1.5x) and largest for terms with significant non-AI uses (diffusion model, 7.7x).
7. **AI research is predominantly open access.** 60.8% of the corpus is published as open access literature, more than double the 28% baseline across all academic fields estimated by Piwowar et al. [11] in 2018. This is likely due to the AI community's preprint culture, where arXiv is widely used.
8. **India is the fastest-growing major producer.** Indian AI research output grew 32.3x from 2013 to 2025 (2,761 to 89,287 papers), a higher growth multiple than both China (15.6x) and the US (8.9x) over the same period.

The methodological contribution of this study is that all results are derived from a single free API (OpenAlex) using abstract-level keyword search. Every count reported in this paper can be independently verified and updated. The corpus definition, search queries, and analysis methods are fully described in Section 2, making the analysis reproducible without access to proprietary databases.

Beyond these eight findings, the sheer volume of AI-related publishing deserves attention. Within this corpus, more papers were published in 2025 alone (788,012) than in the first six years combined (2013-2018, 734,197). More than half of the corpus (58.8%) was published from 2023 onward. If the 2026 trajectory holds, this corpus will exceed 1.6 million papers in a single year.

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